

Financial Econometrics - Exercise 3.18

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Part (a): Fitted Relationship and Visualization

Answer:

- **Mean Calculation:** Based on the fundamental regression property that the fitted line must pass through the point of means, substituting the average income (59.3) into the estimated equation yields a sample mean insurance amount of **236.939**.
- **Intercept (6.855):** This represents the estimated amount of life insurance held when a household's income is zero.
- **Slope (3.880):** This indicates that for every additional \$1,000 in household income, the average amount of life insurance held is expected to increase by **\$3,880**.
- **Point of Means:** The point **(59.3, 236.939)** represents the center of the data sample and lies exactly on the fitted regression line.

```
## a
b1 <- 6.855
b2 <- 3.880
income_mean <- 59.3
insurance_mean <- b1 + b2 * income_mean #計算平均保額
# y_hat = 6.855 + 3.880 * x_hat
insurance_mean
```

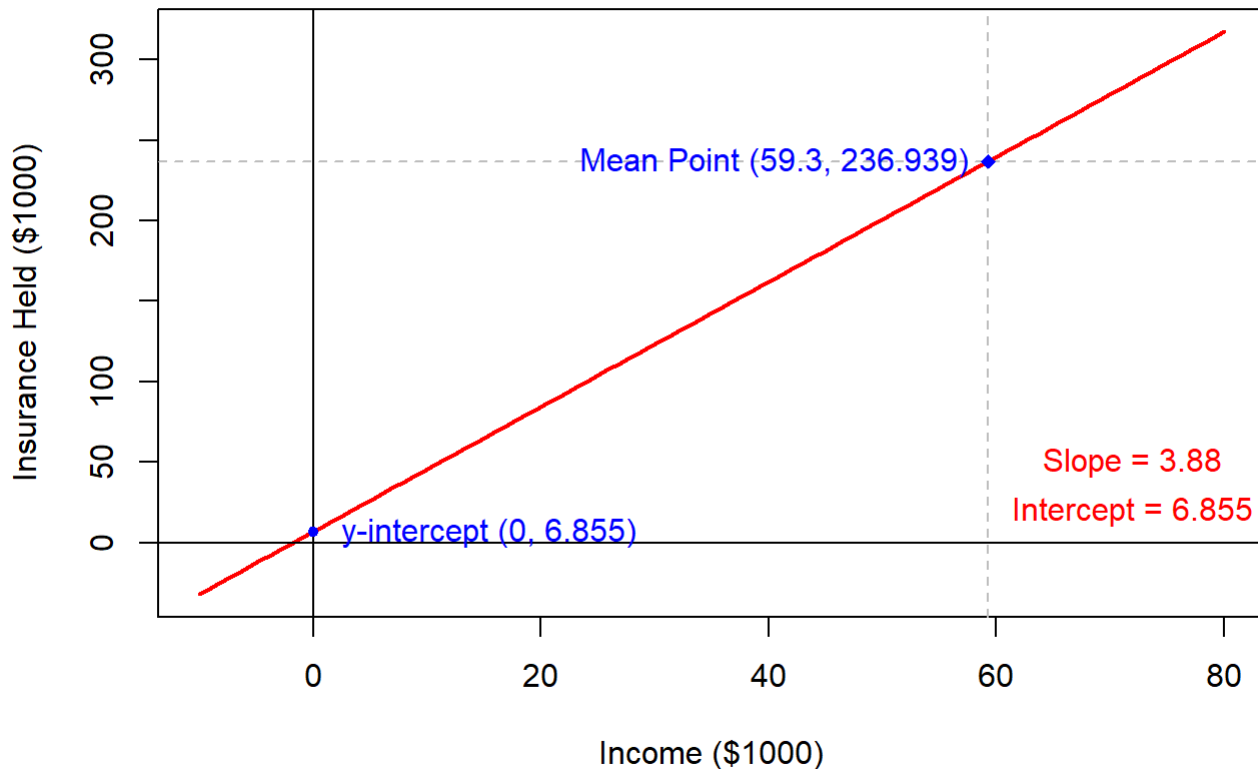
```
## [1] 236.939
```

```
# 繪圖
# 1. 畫擬合線
curve(b1 + b2 * x, from = -10, to = 80, #設定範圍方便顯示
      xlab = "Income ($1000)",
      ylab = "Insurance Held ($1000)",
      main = "Fitted Relationship: Insurance vs Income",
      col = "red", lwd = 2)

# 2. 加入座標軸與輔助虛線，標示出平均點的座標
abline(v = 0, h = 0, lty = 1)
abline(v = income_mean, h = insurance_mean, lty = 2, col = "gray")

# 3. 標註截距點、平均點、截距與斜率
points(0, b1, col = "blue", pch = 20, cex = 1)
text(0, b1, labels = " y-intercept (0, 6.855)", pos = 4, col = "blue")
points(income_mean, insurance_mean, col = "blue", pch = 18, cex = 1)
text(income_mean, insurance_mean, labels = " Mean Point (59.3, 236.939)", pos = 2, col = "blue")
text(72, 50, labels = paste("Slope =", b2), col = "red")
text(72, 20, labels = paste("Intercept =", b1), col = "red")
```

Fitted Relationship: Insurance vs Income



Part (b): Point and Interval Estimates for the Slope

Answer:

- **Point Estimate:** We estimate that the average amount of insurance held increases by **\$3,880** for each additional \$1,000 of household income.
- **95% Interval Estimate:** The 95% confidence interval is **[3.6447, 4.1153]**.
- **Statistical Interpretation for Stockholders:** In repeated sampling, approximately 95% of intervals constructed using this methodology will contain the true parameter value of the increase in life insurance per \$1,000 of income growth. Based on our current sample, this calculated interval is **[\$3,644.7, \$4,115.3]**. This indicates a highly stable and positive correlation between customer wealth and insurance demand.

```
## b
# 1. Point Estimate

point_estimate <- b2
point_estimate
```

```
## [1] 3.88
```

```
# 2. 95% Interval Estimate
se_b2 <- 0.112 # 斜率標準誤
n <- 20
df <- n - 2
t_critical <- qt(0.975, df)

lower_b2 <- b2 - t_critical * se_b2
upper_b2 <- b2 + t_critical * se_b2

# 3. 輸出結果
cat("Point Estimate: ", point_estimate, "\n")
```

```
## Point Estimate: 3.88
```

```
cat("95% Interval Estimate: [", round(lower_b2, 4), ", ", round(upper_b2, 4), "]\n")
```

```
## 95% Interval Estimate: [ 3.6447 , 4.1153 ]
```

Part (c): 99% Interval Estimate for Expected Insurance at \$100,000 Income

Answer:

- **Point Estimate:** For a household with an income of \$100,000, the expected insurance held is estimated to be **\$394.855**.
- **99% Interval Estimate:** The 99% confidence interval for this expected value is **[378.8936, 410.8164]**.
- **Statistical Interpretation:** In repeated sampling, approximately 99% of intervals constructed in this manner will contain the true expected insurance amount for a household with \$100,000 income. This interval accounts for the negative covariance between the intercept and slope, which enhances the precision of the estimate.

```
## c
# 1. 定義已知數值
x0 <- 100 # 收入 100,000 / 1,000
se_b1 <- 7.383 # 截距標準誤
cov_b1b2 <- -0.746 # 截距與斜率的共變異數

# 2. Point Estimate
y0_hat <- b1 + b2 * x0

# 3. 計算預測值的變異數與標準誤
## Var(y_hat) = Var(b1) + 2 * x0 * Cov(b1, b2) + x0^2 * Var(b2)
var_y0 <- (se_b1^2) + (2 * x0 * cov_b1b2) + (x0^2 * se_b2^2)
se_y0 <- sqrt(var_y0)

# 4. 找出 99% t 臨界值 (df = 18, 雙尾各 0.005)
t_critical_99 <- qt(0.995, df)

# 5. 計算 99% 信賴區間
lower_c <- y0_hat - t_critical_99 * se_y0
upper_c <- y0_hat + t_critical_99 * se_y0

# 6. 輸出結果
cat("Point Estimate at Income=100: ", y0_hat, "\n")
```

```
## Point Estimate at Income=100: 394.855
```

```
cat("99% Interval Estimate: [", round(lower_c, 4), ",", round(upper_c, 4), "]\n\n")
```

```
## 99% Interval Estimate: [ 378.8936 , 410.8164 ]
```

Part (d): Hypothesis Test for Slope (Beta2 = 5)

Answer:

- **Hypothesis:** We test the Null Hypothesis ($H_0: \text{Beta2} = 5$) against the Alternative Hypothesis ($H_1: \text{Beta2}$ is not equal to 5).
- **Test Statistic:** The calculated t-statistic is **-10.0**.
- **Conclusion:** Since the absolute value of the t-statistic (10.0) exceeds the critical value (2.101) at the 5% significance level, we **reject the null hypothesis**. The empirical data does not support the claim that the slope is 5.

```
## d
# 1. 定義t檢定參數
b2_claim <- 5 # 董事會成員主張的值
se_b2 <- 0.112 # 斜率標準誤 (來自題目)
alpha <- 0.05 # 顯著水準
n <- 20
df <- n - 2

# 2. 計算t檢定值
t_stat <- (b2 - b2_claim) / se_b2

# 3. 找出臨界值與拒絕域
t_critical_d <- qt(1 - alpha/2, df)

# 4. 顯示結果
cat("Test Statistic (t): ", t_stat, "\n")
```

```
## Test Statistic (t): -10
```

```
cat("Critical Value (tc): ", t_critical_d, "\n")
```

```
## Critical Value (tc): 2.100922
```

```
# 5. 判斷是否拒絕 H0
if (abs(t_stat) > t_critical_d) {
  cat("Result: Reject H0. The data do NOT support the claim.\n")
} else {
  cat("Result: Do not reject H0. The data support the claim.\n")
}
```

```
## Result: Reject H0. The data do NOT support the claim.
```

Part (e): Hypothesis Test for Slope (Beta2 = 1)

Answer:

- **Hypothesis:** We test the Null Hypothesis ($H_0: \text{Beta2} = 1$) against the Alternative Hypothesis ($H_1: \text{Beta2} > 1$).
- **Test Statistic:** The calculated t-statistic is **25.714**.
- **Conclusion:** At the 1% level of significance, the t-statistic far exceeds the one-tailed critical value (2.552). We **reject the null hypothesis** and conclude that the amount of life insurance held increases significantly more than the corresponding increase in income.

```
## e
# 1. 定義單尾檢定參數
b2_null <- 1      # 虛無假設的主張值
alpha_e <- 0.01   # 1% 顯著水準

# 2. 計算t檢定值
t_stat_e <- (b2 - b2_null) / se_b2
# (估計值 3.88 - 主張值 1) / 標準誤 0.112

# 3. 找出單尾檢定臨界值
t_critical_e <- qt(1 - alpha_e, df)
# qt(0.99, df) 代表右邊留下 1% 的面積

# 4. 顯示結果
cat("Test Statistic (t): ", t_stat_e, "\n")
```

```
## Test Statistic (t):  25.71429
```

```
cat("Critical Value (tc) for 1% one-tail: ", t_critical_e, "\n")
```

```
## Critical Value (tc) for 1% one-tail:  2.55238
```

```
# 5. 判斷是否拒絕 H0
if (t_stat_e > t_critical_e) {
  cat("Result: Reject H0. The slope is significantly larger than 1.\n")
} else {
  cat("Result: Do not reject H0. The slope is not significantly larger than 1.\n")
}
```

```
## Result: Reject H0. The slope is significantly larger than 1.
```